

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING) Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Application-Based Questions (3 Questions)**

Question 1

Given

Section A:

- Red = 5
- Blue = 3
- Black = 2
- Total = 10

Section B:

- Red = 4
- Blue = 6
- Black = 5
- Total = 15

Section C:

- Red = 7
- Blue = 2
- Black = 6
- Total = 15

Selection probability:

Section B is twice as likely to be selected.

Hence,

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

A blue car is selected.

Find

$$P(C \mid \text{Blue})$$

Algorithm Used

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

where

- A = Car came from Section C
- B = Selected car is Blue

Step 1: Probability of Blue car from each section

For Section A

$$P(\text{Blue} \mid A) = \frac{3}{10} = 0.3$$

For Section B

$$P(\text{Blue} \mid B) = \frac{6}{15} = 0.4$$

For Section C

$$P(\text{Blue} \mid C) = \frac{2}{15} = 0.1333$$

Step 2: Total probability of selecting a blue car

Using Total Probability Rule,

$$P(\text{Blue}) = P(A)P(\text{Blue} \mid A) + P(B)P(\text{Blue} \mid B) + P(C)P(\text{Blue} \mid C)$$

Substitute the values,

$$= \frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{6}{15} + \frac{1}{4} \times \frac{2}{15}$$

$$\begin{aligned}
 &= 0.075 + 0.20 + 0.0333 \\
 &= 0.3083
 \end{aligned}$$

Step 3: Apply Bayes' Theorem

$$P(C \mid \text{Blue}) = \frac{P(C)P(\text{Blue} \mid C)}{P(\text{Blue})}$$

Substitute values,

$$\begin{aligned}
 &= \frac{\frac{1}{4} \times \frac{2}{15}}{0.3083} \\
 &= \frac{0.0333}{0.3083} \\
 &= 0.108
 \end{aligned}$$

Final Answer

$$\boxed{P(C \mid \text{Blue}) = 0.108}$$

Therefore, the probability that the selected blue car came from **Section C** is

$$\boxed{10.8\%}$$

Question 2

Given Data

Epoch Training Loss Validation Loss

1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

Logic

A good machine learning model should show

- Training loss ↓
- Validation loss ↓

When

- Training loss continues decreasing
- Validation loss starts increasing

the model begins **overfitting**.

(i) Identify the epoch where overfitting starts

Observe the table.

Validation loss

2.4 → 2.2 → 2.0 → **2.1**

The validation loss reaches its minimum at Epoch 3 and starts increasing from Epoch 4.

Training loss continues decreasing.

Hence,

Answer

Overfitting begins at Epoch 4

(ii) Two techniques to reduce overfitting

1. Early Stopping

Stop training when validation loss starts increasing.

In this case,

Training should stop after Epoch 3.

2. Regularization

Apply

- L2 Regularization
- Dropout

These methods prevent the model from memorizing the training data and improve generalization.

(iii) Effect of overfitting

Overfitting causes the model to memorize the training dataset instead of learning general patterns. As a result, the training loss becomes very low while the validation loss increases. The model performs well on training data but poorly on unseen test data, leading to poor generalization and reduced prediction accuracy.

Question 3

Given Data

Epoch	Training Accuracy	Validation Accuracy
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1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

Logic

A good model should have

- Training Accuracy $\uparrow$
- Validation Accuracy $\uparrow$

When

Training Accuracy continues increasing but Validation Accuracy starts decreasing, the model begins overfitting.

(i) Identify the epoch where overfitting starts

Validation Accuracy

58 $\rightarrow$ 63 $\rightarrow$ 68 $\rightarrow$ 72 $\rightarrow$ **73** $\rightarrow$ **72**

Maximum validation accuracy occurs at Epoch 5.

After Epoch 5, validation accuracy decreases while training accuracy continues increasing.

Hence,

Answer

Overfitting begins at Epoch 6

(ii) Why does validation accuracy decrease while training accuracy increases?

The model starts memorizing the training data rather than learning general patterns. It performs exceptionally well on the training dataset, causing training accuracy to increase. However, because it has learned noise and specific details of the training data, it cannot generalize effectively to unseen validation data. As a result, the validation accuracy decreases even though the training accuracy continues to improve.

(iii) Two methods to improve validation performance

Method 1

Early Stopping

Stop training at the epoch where validation accuracy is highest (Epoch 5).

Method 2

Regularization or Data Augmentation

Use techniques such as Dropout, L2 Regularization, or Data Augmentation to reduce overfitting and improve the model's ability to generalize.

Conclusion

- Q1: Using Bayes' Theorem, the probability that the blue car came from Section C is 0.108 (10.8%).
- Q2: The model begins to overfit at Epoch 4 because validation loss starts increasing while training loss continues to decrease.
- Q3: The model begins to overfit at Epoch 6 because validation accuracy decreases while training accuracy continues to increase.

These solutions include the required algorithm (Bayes' theorem), calculations, logical analysis, and case-based explanations, matching the assessment instructions.